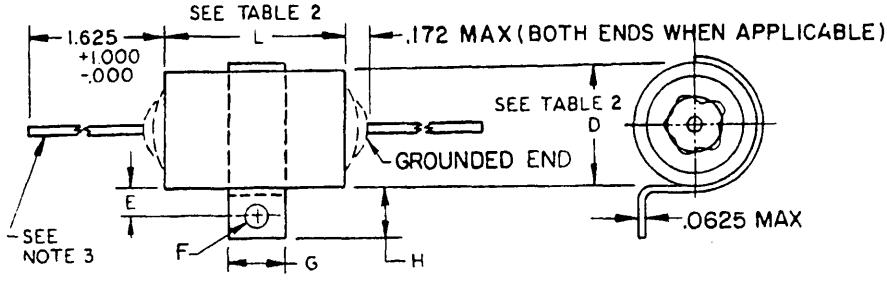
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, PLASTIC (OR PAPER-PLASTIC) DIELECTRIC, AXIAL-WIRE TERMINAL, TANGENTIAL RETAINER, TUBULAR (UNINSULATED) (HERMETICALLY SEALED IN METAL CASES) ESTABLISHED RELIABILITY STYLE CQR12

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-19978.



INCHES	MM	INCHES	MM	INCHES	MM
.001	.03	.0625	1.59	.312	7.92
.004	.10	.144	3.66	.400	10.16
.005	.13	.156	3.96	.438	11.13
.025	.64	.172	4.37	.500	12.70
.031	.79	.188	4.78	.562	14.27
.032	.81	.235	5.97	1.000	25.40
.062	1.57	.250	6.35	1.625	41.28

Case dimension D	H±.062	G±.062	E±.031	F±.005
Less than .562	.312	.250	.188	.144
.562 and up	.438	.500	.250	.156

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Leads shall be of tinned solid wire, .025(No. 22 AWG) for cases .235 and .312 in diameter; and .032(No. 20 AWG) for cases .400 diameter and above. Tolerance on all lead wire diameters shall be +.004, -.001.

FIGURE 1. Style CQR12 capacitors.

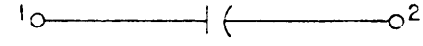
FSC 5910

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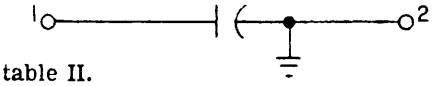
REQUIREMENTS:

Design and construction:
Dimensions and configuration - See figure 1 and table II.
Case material - Nonmagnetic (end seal may be of magnetic material).
Dielectric material - Paper and polyethelene terephthalate (K).
Terminals - Axial-wire lead (see figure 1).

Circuit diagram - 1



3



Rated voltage - See table II.
Rated temperature - -65° to +125° C.
Capacitance (Cap.) (nom) - See table II.
Capacitance tolerance - See table II.
Dissipation factor (DF) (max) - 1.0 percent.
Failure rate level - In accordance with MIL-C-19978.
Reliability - In accordance with MIL-C-19978

Burn-in - In accordance with MIL-C-19978.

Radiographic inspection: In accordance with MIL-C-19978.

Seal: Method 112 of MIL-STD-202, test condition letter - A.

Barometric pressure: In accordance with MIL-C-19978.
Test points:
Circuit diagram 1 - Between terminals and case.
Circuit diagram 3 - Between ungrounded terminal and case.

Insulation resistance (IR):
Terminal to terminal - See table I.

TABLE I. Terminal-to-terminal insulation resistance.

Capacitance rating	Minimum insulation resistance
Characteristic K	
	At 25° C
0 to 0.6 microfarad - - - - - - - - - -	25,000 megohms
Greater than 0.6 microfarad - - - - -	15,000 megohm-microfarads 1/
	At 125° C
0 to 0.08 microfarad - - - - - - - - -	250 megohms
Greater than 0.08 microfarad- - - - -	20 megohm-microfarads 1/

1/ Product obtained by multiplying the capacitance in microfarads by the insulation resistance in megohms.

Terminal to case - Greater than 10,000 megohms.

Vibration, high frequency: Method 204 of MIL-STD-202, test condition B, with the following exception:
Direction and duration of motion - 4 hours in each of two mutually perpendicular directions (total of 8 hours), one parallel and the other perpendicular to the cylindrical axis.

Salt spray: In accordance with MIL-C-19978.

Solderability: In accordance with MIL-C-19978.

Terminal strength: Method 211 of MIL-STD-202, test condition letter - D.

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TABLE II. Style CQR12 capacitors.

Part number 1/	DC voltage rating	Capacitance rating	Capacitance tolerance	Failure rate levels	Case dimensions			
					Circuit 1		Circuit 3	
					L ± .031	D +.015 -.005	L ± .031	D +.015 -.005
	Volts	µF			Inches	Inches	Inches	Inches
CQR12A-KC392-3	200	.0039	G, J, K	L, M, P, R, S	.750	.235	.688	.235
CQR12A-KC472-3	200	.0047	G, J, K		.750	.235	.688	.235
CQR12A-KC562-3	200	.0056	G, J, K		.750	.235	.688	.235
CQR12A-KC682-3	200	.0068	G, J, K		.750	.235	.688	.235
CQR12A-KC183-3	200	.018	G, J, K		.875	.312	.812	.312
CQR12A-KC223-3	200	.022	G, J, K		.875	.312	.812	.312
CQR12A-KC273-3	200	.027	G, J, K		.875	.312	.812	.312
CQR12A-KC333-3	200	.033	G, J, K		.875	.312	.812	.312
CQR12A-KC393-3	200	.039	G, J, K		.875	.400	.812	.400
CQR12A-KC473-3	200	.047	G, J, K		.875	.400	.812	.400
CQR12A-KC563-3	200	.056	G, J, K		.875	.400	.812	.400
CQR12A-KC683-3	200	.068	G, J, K		.875	.400	.812	.400
CQR12A-KC823-3	200	.082	G, J, K		1.125	.400	1.062	.400
CQR12A-KC104-3	200	.10	G, J, K		1.125	.400	1.062	.400
CQR12A-KC124-3	200	.12	G, J, K		1.375	.400	1.312	.400
CQR12A-KC154-3	200	.15	G, J, K		1.375	.400	1.312	.400
CQR12A-KC184-3	200	.18	G, J, K		1.125	.562	1.062	.562
CQR12A-KC224-3	200	.22	G, J, K		1.125	.562	1.062	.562
CQR12A-KC274-3	200	.27	G, J, K		1.375	.562	1.312	.562
CQR12A-KC334-3	200	.33	G, J, K		1.375	.562	1.312	.562
CQR12A-KC394-3	200	.39	G, J, K		1.625	.562	1.562	.562
CQR12A-KC474-3	200	.47	G, J, K		1.625	.562	1.562	.562
CQR12A-KC564-3	200	.56	G, J, K		1.625	.670	1.562	.670
CQR12A-KC684-3	200	.68	G, J, K		1.625	.670	1.562	.670
CQR12A-KC824-3	200	.82	G, J, K		2.125	.750	2.062	.750
CQR12A-KC105-3	200	1.000	G, J, K		2.125	.750	2.062	.750
CQR12A-KE272-3	400	.0027	G, J, K		.750	.235	.688	.235
CQR12A-KE332-3	400	.0033	G, J, K		.750	.235	.688	.235
CQR12A-KE123-3	400	.012	G, J, K		.875	.312	.812	.312
CQR12A-KE153-3	400	.015	G, J, K		.875	.312	.812	.312
CQR12A-KE273-3	400	.027	G, J, K		.875	.400	.812	.400
CQR12A-KE333-3	400	.033	G, J, K		.875	.400	.812	.400
CQR12A-KE393-3	400	.039	G, J, K		1.125	.400	1.062	.400
CQR12A-KE473-3	400	.047	G, J, K		1.125	.400	1.062	.400
CQR12A-KE563-3	400	.056	G, J, K		1.375	.400	1.312	.400
CQR12A-KE683-3	400	.068	G, J, K		1.375	.400	1.312	.400
CQR12A-KE823-3	400	.082	G, J, K		1.125	.562	1.062	.562
CQR12A-KE104-3	400	.10	G, J, K		1.125	.562	1.062	.562
CQR12A-KE124-3	400	.12	G, J, K		1.375	.562	1.312	.562
CQR12A-KE154-3	400	.15	G, J, K		1.375	.562	1.312	.562
CQR12A-KE184-3	400	.18	G, J, K		1.625	.562	1.562	.562
CQR12A-KE224-3	400	.22	G, J, K		1.625	.562	1.562	.562
CQR12A-KE274-3	400	.27	G, J, K		1.625	.670	1.562	.670
CQR12A-KE334-3	400	.33	G, J, K		1.625	.670	1.562	.670
CQR12A-KE394-3	400	.39	G, J, K		2.125	.750	2.062	.750
CQR12A-KE474-3	400	.47	G, J, K	L, M, P, R, S	2.125	.750	2.062	.750

See footnote at end of table.

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TABLE II. Style CQR12 capacitors. -Continued

Part number 1/	DC voltage rating	Capacitance rating	Capacitance tolerance	Failure rate levels	Case dimensions			
					Circuit 1		Circuit 3	
					L±.031	D +.015 -.005	L±.031	D +.015 -.005
	Volts	µF			Inches	Inches	Inches	Inches
CQR12A-KF102-3 -	600	.0010	G, J, K	M, P, R, S	.750	.235	.688	.235
CQR12A-KF122-3 -	600	.0012	G, J, K		.750	.235	.688	.235
CQR12A-KF152-3 -	600	.0015	G, J, K		.750	.235	.688	.236
CQR12A-KF182-3 -	600	.0018	G, J, K		.750	.235	.688	.235
CQR12A-KF222-3 -	600	.0022	G, J, K		.750	.235	.688	.235
CQR12A-KF272-3 -	600	.0027	G, J, K		.875	.312	.812	.312
CQR12A-KF332-3 -	600	.0033	G, J, K		.875	.312	.812	.312
CQR12A-KF392-3 -	600	.0039	G, J, K		.875	.312	.812	.312
CQR12A-KF472-3 -	600	.0047	G, J, K		.875	.312	.812	.312
CQR12A-KF562-3 -	600	.0056	G, J, K		.875	.312	.812	.312
CQR12A-KF682-3 -	600	.0068	G, J, K		.875	.312	.812	.312
CQR12A-KF822-3 -	600	.0082	G, J, K		.875	.312	.812	.312
CQR12A-KF103-3 -	600	.010	G, J, K		.875	.312	.812	.312
CQR12A-KF123-3 -	600	.012	G, J, K		.875	.400	.812	.400
CQR12A-KF153-3 -	600	.015	G, J, K		.875	.400	.812	.400
CQR12A-KF183-3 -	600	.018	G, J, K		.875	.400	.812	.400
CQR12A-KF223-3 -	600	.022	G, J, K		.875	.400	.812	.400
CQR12A-KF273-3 -	600	.027	G, J, K		1.125	.400	1.062	.400
CQR12A-KF333-3 -	600	.033	G, J, K		1.125	.400	1.062	.400
CQR12A-KF393-3 -	600	.039	G, J, K		1.375	.400	1.312	.400
CQR12A-KF473-3 -	600	.047	G, J, K		1.375	.400	1.312	.400
CQR12A-KF563-3 -	600	.056	G, J, K		1.125	.562	1.062	.562
CQR12A-KF683-3 -	600	.068	G, J, K		1.125	.562	1.062	.562
CQR12A-KF823-3 -	600	.082	G, J, K		1.375	.562	1.312	.562
CQR12A-KF104-3 -	600	.10	G, J, K		1.375	.562	1.312	.562
CQR12A-KF124-3 -	600	.12	G, J, K		1.625	.562	1.562	.562
CQR12A-KF154-3 -	600	.15	G, J, K		1.625	.562	1.562	.562
CQR12A-KF184-3 -	600	.18	G, J, K		1.625	.670	1.562	.670
CQR12A-KF224-3 -	600	.22	G, J, K		1.625	.670	1.562	.670
CQR12A-KF274-3 -	600	.27	G, J, K		2.125	.750	2.062	.750
CQR12A-KF334-3 -	600	.33	G, J, K		2.125	.750	2.062	.750
CQR12A-KF394-3 -	600	.39	G, J, K		2.375	.750	2.312	.750
CQR12A-KF474-3 -	600	.47	G, J, K		2.375	.750	2.312	.750
CQR12A-KG102-3 -	1000	.0010	G, J, K		.875	.400	.812	.400
CQR12A-KG122-3 -	1000	.0012	G, J, K		.875	.400	.812	.400
CQR12A-KG152-3 -	1000	.0015	G, J, K		.875	.400	.812	.400
CQR12A-KG182-3 -	1000	.0018	G, J, K		.875	.400	.812	.400
CQR12A-KG222-3 -	1000	.0022	G, J, K		.875	.400	.812	.400
CQR12A-KG272-3 -	1000	.0027	G, J, K		.875	.400	.812	.400
CQR12A-KG332-3 -	1000	.0033	G, J, K		.875	.400	.812	.400
CQR12A-KG392-3 -	1000	.0039	G, J, K		.875	.400	.812	.400
CQR12A-KG472-3 -	1000	.0047	G, J, K		.875	.400	.812	.400
CQR12A-KG562-3 -	1000	.0056	G, J, K		.875	.400	.812	.400
CQR12A-KG682-3 -	1000	.0068	G, J, K		.875	.400	.812	.400
CQR12A-KG822-3 -	1000	.0082	G, J, K		.875	.400	.812	.400
CQR12A-KG103-3 -	1000	.010	G, J, K	M, P, R, S	.875	.400	.812	.400

See footnote at end of table.

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TABLE II. Style CQR12 capacitors. -Continued

Part number 1/	DC voltage rating	Capacitance rating	Capacitance tolerance	Failure rate level	Case dimensions			
					Circuit 1		Circuit 3	
					L±.031	D +.015 -.005	L±.031	D +.015 -.005
	Volts	μF			Inches	Inches	Inches	Inches
CQR12A-KG123-3-	1000	.012	G, J, K	M, P, R, S	1.125	.400	1.062	.400
CQR12A-KG153-3-	1000	.015	G, J, K		1.125	.400	1.062	.400
CQR12A-KG183-3-	1000	.018	G, J, K		1.375	.400	1.312	.400
CQR12A-KG223-3-	1000	.022	G, J, K		1.375	.400	1.312	.400
CQR12A-KG273-3-	1000	.027	G, J, K		1.125	.562	1.062	.562
CQR12A-KG333-3-	1000	.033	G, J, K		1.125	.562	1.062	.562
CQR12A-KG393-3-	1000	.039	G, J, K		1.375	.562	1.312	.562
CQR12A-KG473-3-	1000	.047	G, J, K		1.375	.562	1.312	.562
CQR12A-KG563-3-	1000	.056	G, J, K		1.625	.562	1.562	.562
CQR12A-KG683-3-	1000	.068	G, J, K		1.625	.562	1.562	.562
CQR12A-KG823-3-	1000	.082	G, J, K		1.625	.670	1.562	.670
CQR12A-KG104-3-	1000	.10	G, J, K		1.625	.670	1.562	.670
CQR12A-KG124-3-	1000	.12	G, J, K		1.875	.670	1.812	.670
CQR12A-KG154-3-	1000	.15	G, J, K		1.875	.670	1.812	.670
CQR12A-KG184-3-	1000	.18	G, J, K		2.125	.750	2.062	.750
CQR12A-KG224-3-	1000	.22	G, J, K		2.125	.750	2.062	.750
CQR12A-KG274-3-	1000	.27	G, J, K		2.125	1.0	2.062	1.000
CQR12A-KG334-3-	1000	.33	G, J, K		2.125	1.0	2.062	1.000
CQR12A-KG394-3-	1000	.39	G, J, K		2.375	1.0	2.312	1.000
CQR12A-KG474-3-	1000	.47	G, J, K	M, P, R, S	2.375	1.0	2.312	1.000

1/ Complete part number shall include additional symbols to indicate circuit, capacitance tolerance, and failure rate level, as applicable.

TABLE III. Millimeter equivalents of decimal inches.

Inches	mm	Inches	mm	Inches	mm
0.005	0.13	0.670	17.02	1.125	28.58
0.015	0.38	0.688	17.48	1.312	33.32
0.031	0.79	0.750	19.05	1.375	34.93
0.235	5.97	0.812	20.62	1.562	39.67
0.312	7.92	0.875	22.23	1.625	41.28
0.400	10.16	1.000	25.40	2.062	52.37
0.562	14.27	1.062	26.97	2.125	53.98
				2.312	58.72
				2.375	60.33

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Stability at low and high temperature:
Low temperature:
Test temperature - $-65^{\circ}\ ^{+0^{\circ}}_{-5^{\circ}}$ C.
Capacitance change (max) - -10 percent.
High temperature:
Test temperature - $+125^{\circ}\ ^{+5^{\circ}}_{-0^{\circ}}$ C.
Capacitance change (max) - +10 percent.

Life:
Capacitance change (max) - ±5 percent of initial measured value.

Resistance to soldering heat: In accordance with MIL-C-19978.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - MI, MU
Navy - AS
Air Force - 11, 17
DSA - ES

User activities:
Army - ME, AT, AV
Navy - MC, AS, OS, SH
Air Force -

Preparing activity:
Air Force - 80

Agent:
DSA - ES

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	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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